

MENTORING OF ENGINEERING STUDENTS BY INAE FELLOWS/ INAE YOUNG ASSOCIATES

REPORT BY ENGINEERING STUDENT

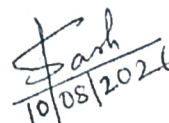
(To be submitted soon after completion of the program, in no case later than 2 months)

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7. **Title of the Research Project/Work undertaken:** Implementation and Performance Analysis of One- and Two-Way DV-QKD MIMO FSO Communication Systems Using SeQUeNCe Simulator
8. Details of the Work Carried Out: Provided in the Annexure
9. Likely Benefits of the Outcome of the Research Work:
 (a) to the student: Provided in the Annexure
 (b) to the state of knowledge in the field): Provided in the Annexure
10. Future Work and Scope: Provided in the Annexure
11. Suggestions, if any, for Improvement of the Scheme: NA


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10/08/2026

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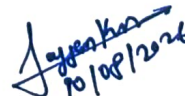
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Title: Implementation and Performance Analysis of One- and Two-Way DV-QKD MIMO FSO Communication Systems Using SeQUeNCe Simulator

1. Introduction

Secure communication has become one of the most important requirements in today's digital world, especially with the growth of 6G networks, IoT devices, and non-terrestrial systems like satellites and high-altitude platforms [1]. Free-space optical (FSO) communication has emerged as a strong candidate to support these networks because it can provide high-speed, license-free, and interference-resistant links over long distances [2]. However, even though FSO links use narrow optical beams which make eavesdropping difficult, they are still not completely secure, especially in dynamic outdoor environments where an eavesdropper can exploit misalignment or atmospheric effects [3].

Among different QKD schemes, discrete-variable QKD (DV-QKD) is widely used because it is compatible with standard photonic hardware [4], [5]. In DV-QKD, weak coherent pulses (WCPs) are commonly used instead of ideal single-photon sources, but this introduces a vulnerability known as the photon-number-splitting (PNS) attack, where an eavesdropper can extract information from multi-photon pulses without being detected. The decoy-state method was introduced to solve this problem by helping the legitimate users estimate the single-photon contribution accurately and detect the presence of an eavesdropper [6]. The one-way BB84 protocol with decoy states is a well-established approach for secure key exchange [7]. Later, a two-way protocol called LM05 was proposed [8].

The motivation of this project is to implement the system described in the above paper through simulation and to verify whether the analytical trends reported in the paper (such as the crossover distance between one-way and two-way protocols) are actually observed when the protocols are simulated using a quantum network simulator.

The main objectives of this work are:

1. To implement the BB84 (one-way) and LM05 (two-way) DV-QKD protocols using the SeQUeNCe simulator.
2. To model the MIMO FSO channel, including beam spreading, atmospheric attenuation, and turbulence-induced fading, using the SVD-based approach given in the paper.
3. To evaluate the SKR and QBER performance of both protocols for different MIMO configurations and transmission distances.
4. To compare the simulation results with the theoretical results of the reference paper and to identify any differences or new observations found during the implementation.

2. System Model

The system considered in this project consists of two legitimate users, Alice and Bob, who want to exchange a secret key over a MIMO FSO channel. Each user has NT laser sources (for transmission) and NR photodetectors (for reception). Depending on the protocol used, either Alice sends the quantum states to Bob (one-way, BB84), or Bob sends the state first, and Alice sends them back after encoding (two-way, LM05).

2.1 MIMO FSO Channel Model

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The optical link between Alice and Bob is affected by diffraction, beam spreading, and turbulence-induced misalignment. The overall channel is represented as an $NR \times NT$ complex gain matrix H , where each element $h(i,j)$ gives the channel gain between the j -th transmit aperture and the i -th receive aperture. This gain depends on the Gaussian beam field distribution at the transmitter, the received field distribution after free-space propagation, and the misalignment between the transmitter and receiver caused by pointing errors and beam wandering.

2.2 One-Way (BB84) Protocol Architecture

In the one-way protocol, **Alice** generates weak coherent pulses (WCPs) from her laser sources, where the mean photon number for each pulse is randomly chosen from a signal state and two decoy states. This randomness is important because it prevents an eavesdropper (Eve) from being able to guess the photon number distribution in advance. The quantum states then travel through the MIMO FSO channel, where **Eve** may try to perform a photon-number-splitting (PNS) attack by extracting information from multi-photon pulses. On reaching **Bob**, threshold detectors are used to detect whether a pulse contains photons or not (they cannot count the exact number of photons). After the key exchange, Alice and Bob use the decoy-state method to estimate the single-photon yield and QBER, which are then used to calculate the final secret key rate.

2.3 Two-Way (LM05) Protocol Architecture

In the two-way protocol, the roles are slightly different. **Bob** starts the process by generating WCPs and sending them to Alice. **Alice** then encodes her classical bit onto the received qubit using the message mode (MM) action, where an identity operation represents 0 and a spin-flip operation represents 1. This mode is preferred over the control mode because it gives higher key generation efficiency. After encoding, Alice sends the qubit back to Bob, who performs threshold detection to extract the key. Since the qubit travels twice through the channel (Bob to Alice, and then Alice to Bob), the overall transmissivity of this protocol depends on the round-trip loss, which includes attenuation on both paths as well as the square of the β_i values. Because of this round-trip nature, Eve gets two chances to perform a PNS attack, once in each direction, which generally makes this protocol more vulnerable to errors at longer distances.

3. Methodology / Work Carried Out

3.1 Tools and Simulation Environment

For this project, I used Python as the main programming language, along with the SeQUeNCe quantum network simulator, which is an open-source discrete-event simulator specifically built for quantum networks.

3.2 Implementation of MIMO FSO Channel

I first implemented a MIMOFSOChannel class which models the channel matrix H described in the system model. This class calculates the field distribution at the transmitter, the received field distribution after propagation, and combines these to build the channel gain matrix for a given number of transmit/receive apertures (NT , NR) and link distance. Singular value decomposition (SVD) is then applied on this matrix to extract the β_i values for each sub-channel. These β_i values, along with atmospheric absorption and turbulence-induced fading (modeled using a lognormal distribution), are used to calculate the final transmissivity of each sub-channel, which is later used by both the BB84 and LM05 protocol implementations.

3.3 Implementation of BB84 Protocol

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For BB84, I implemented a DecoyWCPSource class which generates weak coherent pulses with a randomly selected mean photon number (signal state or one of the two decoy states), following a Poisson distribution, similar to the model described in the reference paper. An EveEntity class was implemented to simulate the photon-number-splitting (PNS) attack, where Eve tries to extract information from multi-photon pulses without disturbing single-photon states. The SeQUeNCe node and protocol structure was used to send these pulses from Alice to Bob, where threshold detection was applied on the receiver side. After the transmission, the QBER and SKR were calculated using the two-decoy-state formulas given in the paper, and the results were plotted for different MIMO configurations and distances.

3.4 Implementation of LM05 Protocol

Since SeQUeNCe does not have a built-in implementation of the LM05 protocol, I had to build a custom protocol class (LM05Protocol) to handle the two-way communication process. In this implementation, Bob generates the WCPs first, and an AliceEncoder class is used to apply either the message mode (identity or spin-flip operation) or control mode on the received qubit, based on the probabilities described in the paper. I also implemented a two-way version of the channel (MIMOFSOChannelTwoWay) to account for the round-trip transmissivity, since the qubit passes through the channel twice (Bob to Alice, and Alice to Bob). An EveEntityTwoWay class was used to simulate the PNS attack in both directions. After the qubit reaches Bob again, sifting is performed by matching Alice's message-mode choices with Bob's detector clicks, and the final QBER and SKR.

3.5 Simulation Parameters

The system parameters used for this simulation were taken directly from the reference paper[1], so that the simulation results could be fairly compared with the analytical results reported there.

4. Results

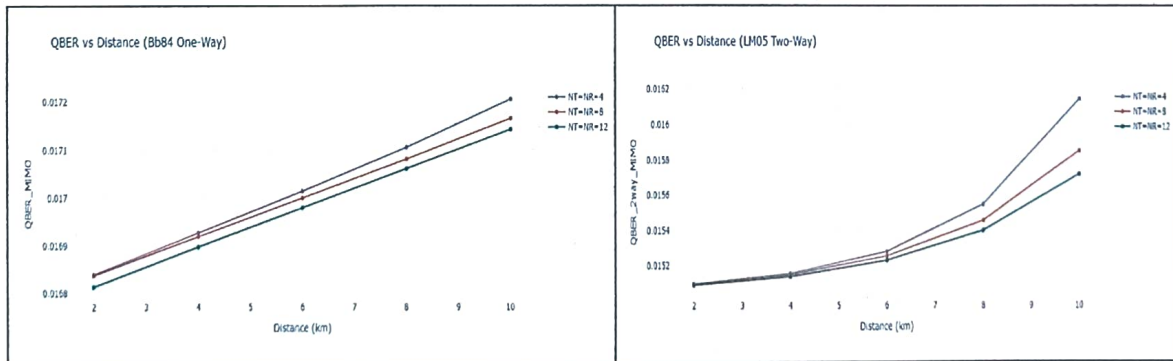
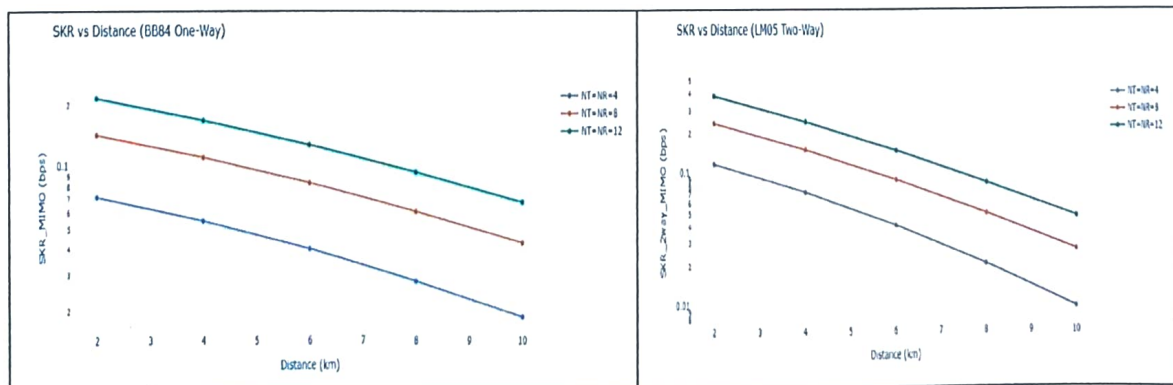


Fig. 5.1(a)

Fig. 5.1(b)



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Fig. 5.2(a)

Fig. 5.2(b)

Fig. 5.1(a) shows the QBER for the BB84 (one-way) protocol, and Fig. 5.1(b) the LM05 (two-way) protocol, both plotted against distance (2-10 km) for $NT = NR = 4, 8, \text{ and } 12$. For BB84, QBER increases gradually with distance, from about 0.0168 at 2 km to about 0.0172 at 10 km. For LM05, QBER also increases with distance, from about 0.0151 at 2 km to about 0.0162 at 10 km.

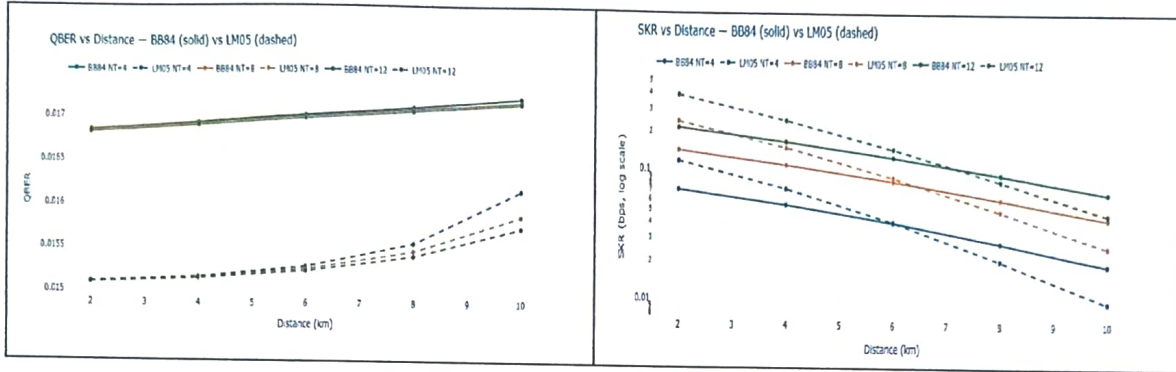
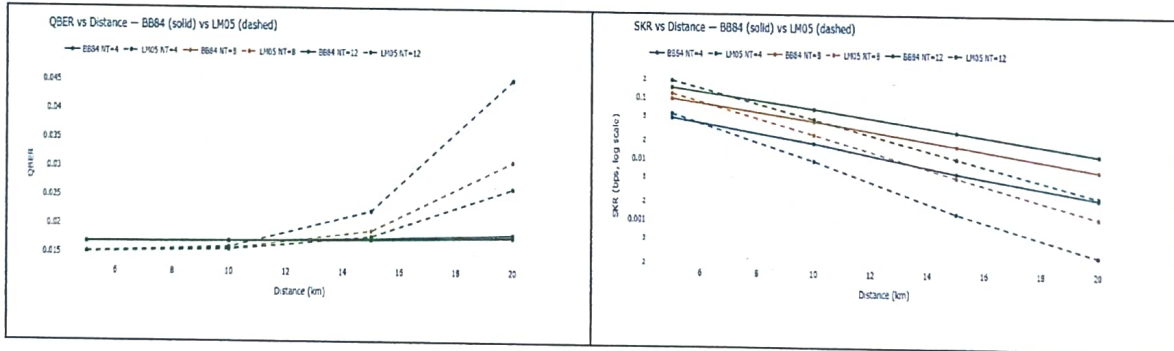


Fig. 5.2(c)

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Fig. 5.2(a) and Fig. 5.2(b) show the SKR (log scale) for BB84 and LM05 respectively. For both protocols, SKR decreases as distance increases, and increases as NT increases, which matches the expected behavior of MIMO gain improving key rate.

To further verify the crossover behavior, the same comparison was extended to a longer distance range (5-20 km).



5. Conclusion and Future Directions

This project implemented and analyzed one-way (BB84) and two-way (LM05) DV-QKD protocols over a MIMO FSO channel using the SeQUeNce quantum network simulator, based on the analytical model of (ICC 2026). The MIMO FSO channel was modeled using SVD-based transmissivity calculation, and both protocols were implemented with decoy-state WCP sources and a PNS-attack eavesdropper model. Simulation results confirmed the paper's central finding that a crossover distance exists beyond which BB84 outperforms LM05 in SKR, while LM05 remains advantageous only for shorter-range links.

Future directions for this work include:

- Gate-level implementation and testing of the QKD protocols on real IBM quantum hardware, to validate the protocol-level results obtained from SeQUeNce against an actual quantum processing backend.

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- Development of a dedicated quantum simulator capable of natively modeling MIMO channel effects (multiple transmit/receive apertures, SVD-based sub-channel decomposition) at the simulator-engine level, rather than relying on custom channel classes layered on top of a SISO-oriented simulator like SeQUeNCe.

6. Likely Benefits of the Outcome of the Research Work:

(a) To the student: This project gave me hands-on experience implementing BB84 and LM05 QKD protocols using the SeQUeNCe simulator, instead of only studying them theoretically. I learned to model a MIMO FSO channel, apply decoy-state methods, simulate an eavesdropper attack, and analyze SKR/QBER trade-offs, which strengthened my practical skills for further work in quantum communication.

(b) To the state of knowledge in the field: This work adds a simulation-based validation of the analytical framework proposed in the reference ICC 2026 paper, using an independent event-driven quantum network simulator. It confirms the paper's core theoretical trends (SKR-distance decay, MIMO gain, crossover between one-way and two-way protocols) while also surfacing simulation-specific findings, such as a shorter observed crossover distance and short-range QBER behavior differences, that are not captured in the paper's closed-form analytical results. This helps bridge the gap between analytical QKD modeling and simulator/hardware-level validation.

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